

AI/ML intern DAY 4

Task 2 — Virtual Try-On AI Models Research

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1. Objective

The objective of this task was to research and understand the working of AI-powered virtual try-on systems used in modern fashion technology and e-commerce platforms.

The task focused on studying:

- body tracking systems
- pose detection
- body segmentation
- clothing overlay methods
- AI-based fashion visualization

The research also explored how computer vision and Artificial Intelligence improve online shopping experiences.

2. Introduction to Virtual Try-On AI Systems

Virtual try-on AI systems are advanced computer vision applications that allow users to digitally wear clothes, shoes, glasses, or accessories before purchasing them online.

These systems use:

- Artificial Intelligence
- Computer Vision
- Pose Detection
- Body Segmentation

- Augmented Reality

to create interactive and realistic virtual fitting experiences.

Virtual try-on technology has become highly important in:

- online fashion shopping
- fashion recommendation systems
- virtual fitting rooms
- AI styling platforms

The main purpose of virtual try-on systems is to help customers visualize products digitally and improve shopping confidence.

3. Working of Virtual Try-On AI Systems

Virtual try-on systems work using multiple AI and computer vision processes.

Step 1 — Human Detection

The AI system first detects the human body using computer vision algorithms.

Technologies commonly used:

- OpenCV
- Deep Learning
- TensorFlow
- Human Detection Models

Step 2 — Pose Detection

Pose detection algorithms identify important body landmarks such as:

- shoulders
- elbows
- wrists
- hips
- knees

- ankles

MediaPipe and OpenPose are commonly used for body landmark detection.

Step 3 — Body Segmentation

The system separates the human body from the background using segmentation algorithms.

This helps:

- identify body shape
 - improve clothing alignment
 - increase fitting accuracy
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Step 4 — Clothing Overlay

Virtual clothing is digitally placed over the user's body using AI alignment techniques.

The system adjusts:

- clothing size
 - body posture
 - clothing position
 - body movement
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Step 5 — Real-Time Visualization

Users can move in front of the camera and view live clothing visualization dynamically.

4. Applications of Virtual Try-On AI Systems

Virtual try-on systems are widely used in:

- online fashion stores
- virtual fitting rooms

- eyewear applications
 - makeup applications
 - footwear shopping platforms
 - AI fashion assistants
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Fashion E-Commerce

Customers can visualize clothes digitally before purchasing products online.

Personalized Shopping

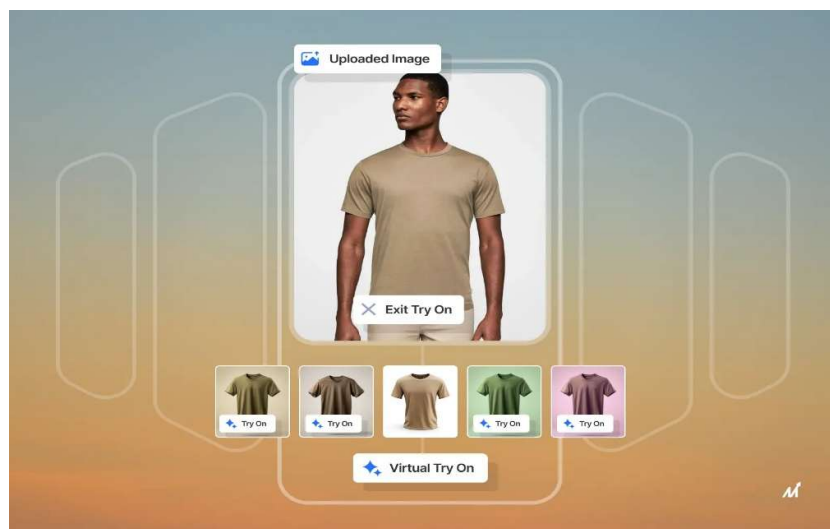
AI systems provide personalized clothing suggestions based on:

- body measurements
 - customer preferences
 - fashion trends
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Virtual Styling

AI systems recommend:

- outfit combinations
- matching accessories
- fashion styles



5. Advantages of Virtual Try-On AI Systems

Virtual try-on systems provide several advantages:

- improved online shopping experience
- reduced product return rates
- accurate clothing size estimation
- increased customer confidence
- better product visualization
- personalized fashion recommendations
- faster shopping decisions

6. Challenges in Virtual Try-On Systems

Challenge	Description	Practical Mitigation
Accurate body measurement	Estimates vary with viewpoint and pose, causing measurement error	Capture known-distance views (calibration), enforce frontal/standardized poses, and use multi-frame averaging
Clothing alignment	Loose or thick clothing hides landmarks and distorts silhouette	Require tight/fit clothing for capture, detect loose clothing and warn user, or use clothing-aware models
Real-time processing	High compute needs cause lag or dropped frames	Resize frames, run a lightweight model (or lower inference rate), use GPU/accelerator, and batch/async processing
Lighting conditions	Poor or uneven lighting reduces landmark visibility and confidence	Ask user to move to well-lit area, apply contrast/denoise preprocessing, and use automatic exposure hints

7. MediaPipe vs OpenPose in Virtual Try-On Systems

Feature	MediaPipe	OpenPose
Speed	Faster	Slower
Accuracy	High	Very High
Real-Time Performance	Excellent	Moderate
Ease of Use	Easy	Complex
Hardware Requirement	Low	High
Best For	Lightweight AI systems	Advanced research applications

8. Learning Outcomes

During this research task, the following concepts were learned:

- virtual try-on workflows
- body segmentation
- AI-based body tracking
- pose estimation systems
- clothing recommendation systems
- computer vision applications in fashion
- MediaPipe and OpenPose comparison
- AI applications in e-commerce

9. Conclusion

Virtual try-on AI systems are transforming the fashion industry by improving online shopping experiences using Artificial Intelligence and computer vision technologies.

These systems combine:

- body tracking
- pose detection
- body segmentation
- augmented reality

to create interactive digital fitting experiences.

The research provided practical understanding of:

- AI fashion technologies
- virtual fitting systems
- pose estimation
- computer vision workflows
- AI-powered shopping systems

The study also highlighted the importance of MediaPipe, OpenPose, and deep learning technologies in modern fashion applications.

